Notebook 2012 — Pagina's 27–31

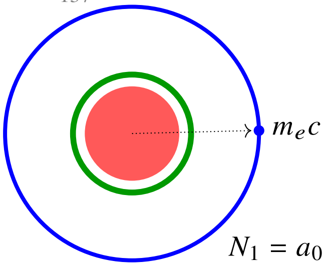
Pagina 26 is op verzoek overgeslagen. De reconstructie hieronder volgt de handgeschreven notities zo letterlijk mogelijk. Waar een symbool of label niet volledig scherp leesbaar is, is de meest waarschijnlijke lezing gekozen.

Pagina 27

Linkerhelft

$$a_0 = \frac{\hbar}{m_e c \alpha}$$
$$= 0{,}52917721092 \cdot 10^{-10} \text{ meter}$$

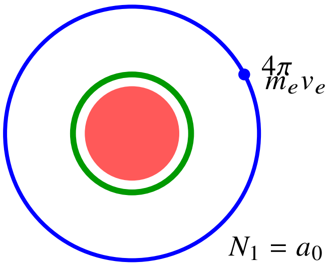
Groene notitie: "Wat is α*"; daarnaast vermoedelijk* $\alpha \approx \frac{1}{137}$.



$$\frac{1.054571726 \cdot 10^{-34}}{9.10938291 \cdot 10^{-31} \cdot 299792458 \cdot 0.00729735256}$$

Rechterhelft

$$h = 4\pi m_e v_e a_0$$
$$= 6{,}62606957 \cdot 10^{-34} \text{ J s}$$



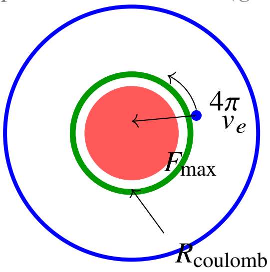
$4\pi \cdot 9.10938291 \cdot 10^{-31} \cdot 1.0938456336 \cdot 10^{6} \cdot 0.52917721092 \cdot 10^{-10}$

Pagina 28

Linkerhelft

$$h = \frac{4\pi F_{\max} R_c^2}{v_e} = 6{,}62606957 \cdot 10^{-34}\ \mathrm{J\,s}$$

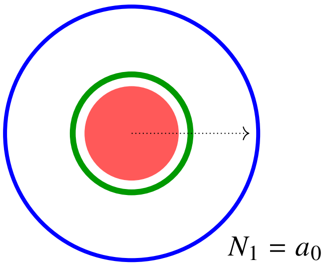
Losse notitie rechtsboven: "Planck is impuls/hoekmoment" (gedeeltelijk onzeker).



$$\frac{4\pi \cdot 29.053507333 \cdot (1.46097017 \cdot 10^{-15})^2}{1.0938456336 \cdot 10^6}$$

Rechterhelft

$$a_0 = \frac{h}{4\pi m_e v_e} = 0{,}52917721 \cdot 10^{-10}$$



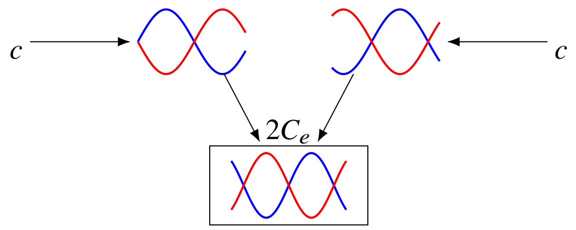
$$\frac{6.62606957 \cdot 10^{-34}}{4\pi \cdot 9.10938291 \cdot 10^{-31} \cdot 1.0938456336 \cdot 10^6}$$

Pagina 29

Linkerhelft

$$
\begin{aligned}
a_0 &= \frac{c^2 R_c}{2C_e} \\
&= 0{,}52917721092 \cdot 10^{-10}
\end{aligned}
$$

$$
\lambda = \frac{2C_e}{c}
$$



De doorgestreepte notitie onderaan is te onduidelijk om betrouwbaar over te nemen.

Rechterhelft

$$
\begin{aligned}
R_c &= \frac{c^2}{a_0\, 2C_e} \\
&= 1{,}46097017 \cdot 10^{-15}
\end{aligned}
$$

$$
\frac{299792458^2}{0.52917721092 \cdot 10^{-10} \cdot 2 \cdot 1.0938456336 \cdot 10^6}
$$

Deze paginahelft bevat verder vrijwel alleen de numerieke invulling.

Pagina 30

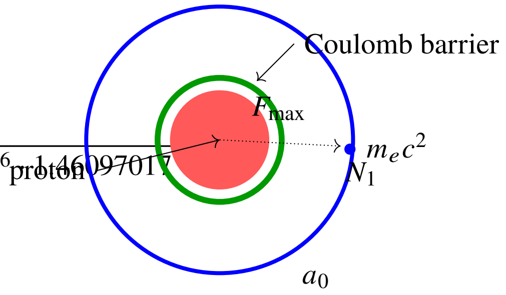
Linkerhelft

$$a_0 = h \cdot \frac{c^2}{8\pi F_{\max} v_e R_c} = 0{,}529177211 \cdot 10^{-10}$$

$$6.62606957 \cdot 10^{-34} \cdot \frac{299792458^2}{8\pi \cdot 29.053507333 \cdot 1.0938456336 \cdot 10^6 \cdot 1.46097017}$$

Rechterhelft

$$a_0 = \frac{F_{\max} R_c}{m_e v_e^2}$$

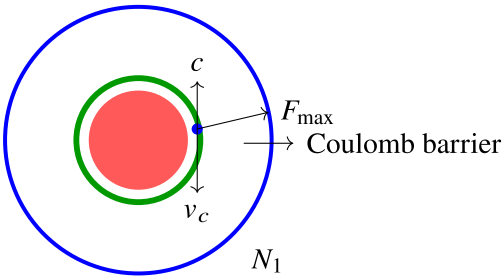


$$\frac{29.053507333 \cdot (1.46097017 \cdot 10^{-15})^2}{9.10938291 \cdot 10^{-31} \cdot (1.0938456336 \cdot 10^6)^2}$$

Pagina 31

Linkerhelft

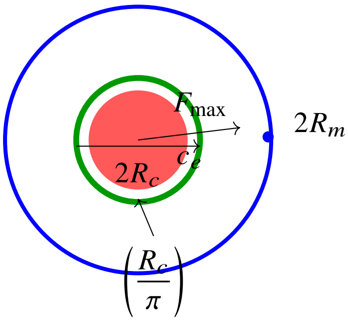
$$
\begin{aligned}
m_e &= \frac{2F_{\max}R_c}{c^2} \\
&= 9{,}10938287 \cdot 10^{-31}
\end{aligned}
$$



$$\frac{2 \cdot 29.053507333 \cdot 1.46097017 \cdot 10^{-15}}{299792458^2}$$

Rechterhelft

$$
\begin{aligned}
v_e &= \left(\frac{R_c}{\pi}\right)\sqrt{\frac{F_{\max}}{2R_{\text{proton}}m_{\text{proton}}}} \\
&= 1{,}0938456336 \cdot 10^{6}
\end{aligned}
$$



$$\left(\frac{1.46097017 \cdot 10^{-15}}{\pi}\right)\sqrt{\frac{29.053507333}{2 \cdot 1.4600300097 \cdot 10^{-15} \cdot 1.672621777}}$$